

The ICT Lounge

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Year 7

Year 8

IGCSE

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Section 7.5: Stage 3b - Testing a System

Once the system has been developed (created), it needs to be **fully tested** to make sure that it is **functioning correctly**.

Testing takes place during the second part of **stage 3**.

During the testing stage, **every part** of the system will be checked in order to **locate** any **errors**. These errors will be **corrected** and then re-tested to ensure that every part of the system does the job it was designed to do.

Key Concepts of this section:

- # Understand what is meant by **systems testing**.
- # Know the **difference** between **unit testing** and **whole-system testing**.
- # Be able to create a **test plan**, using **normal**, **extreme**, **abnormal** and **live** test data.

Stage 3b - Testing

Testing individual modules (parts) of the system

- # When each **module** (part) of the system has been created it must be **tested** to make sure that it **works correctly**.

Examples of modules that should be tested include:

- **Data structures** - do tables hold data correctly?
- **Validation rules** - does the system reject unreasonable/incorrect data?
- **Input screens** - does each form control allows users to enter data correctly?
- **Output screens** - are output results correct, clear and complete?

Testing individual parts (modules) of the system is known as **unit testing**.



Key Words:

Modules, Test Plan, Normal, Extreme, Abnormal, Live Data, Unit Test, Integrated Test

Examples:

- # Any **errors** found will be **corrected** by the person who created the module (a programmer for example).

- # The module that failed the first test will then be **re-tested** to make sure that the error has been **fixed**.

Using a test plan

- # Testing a system involves creating and using a **test plan**.
- # A test plan should be created for each system module should **list** all of the **different tests** that we are going to perform.

A good test plan should be created for every system module and include...

- * A **list** of the tests that are to be performed
- * The **data** to be used in the test
- * The **type of test** -(normal / extreme / abnormal / live)
- * The **expected outcome** of the test
- * The **actual outcome** of the test should be logged (Data accepted / rejected).

Testing with normal, extreme and abnormal data

- # A test plan should always use **four** types of testing data:

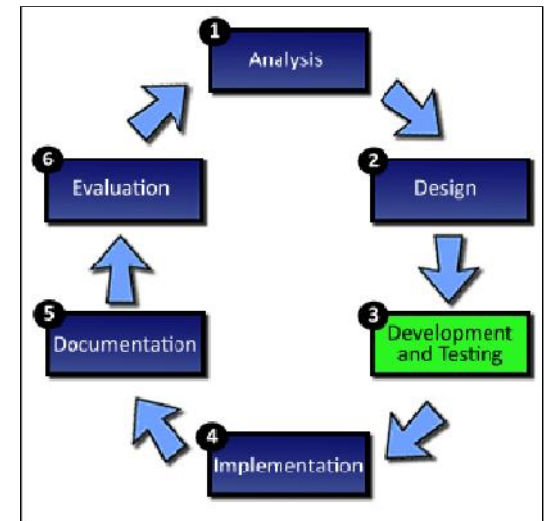
- **Normal data**
- **Extreme data**
- **Abnormal data**
- **Live data**

Scenario:

Imagine we were testing a system module (text box) to make sure that it will **only accept** entries of **numbers between 1 and 5**. The test data would look like the example in the table below:

The first three types of test data would be used to test the system BEFORE it was delivered to the customer.

Type of test data	Description	Example
Normal data	Data which should be accepted and pass the test without any problems. (In our example, this was any number between 1 and 5)	The numbers 1, 2, 3, 4 or 5 should be accepted.



Testing takes place as part of the third stage of the systems lifecycle.



System database modules need to be tested to make sure they hold data correctly.



System validation modules need to be tested to make sure that system errors are rejected.

Extreme data	Data which is on the border of what the system will accept.	Using the same scenario as above, the numbers 1 and 5 would be used to test the borderline data.
Abnormal data	Data that should not be accepted by the system. (In our example, this is any data other than 1, 2, 3, 4 or 5)	Examples of data that should be rejected by the system could be 0, 6, Two, Hello , etc.
Live data (below) is used to test the system AFTER it has been installed into the customer's workplace.		
Type of test data	Description	Example
Live data	Data that is actually used in the customer's company.	Once installed into the customer's workplace, all modules would be tested with real-life data that the company actually uses .

System data entry modules need to be tested to make sure that data can be entered correctly.

System output modules need to be tested to make sure that results are correct.

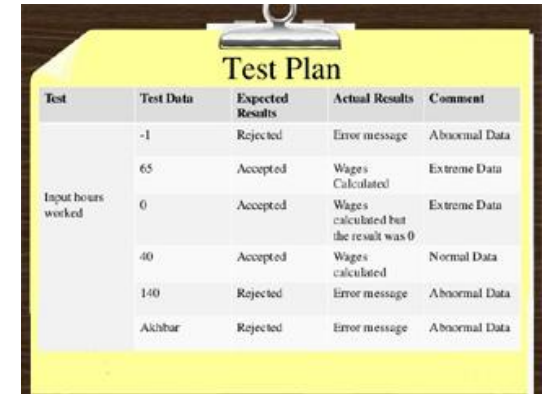
An example test plan

- # The table below demonstrates an **example test plan** that could be used to test a **system module** that will be used to accept the age of a driving test candidate.

The cell should **accept** ages within the range of **17 - 70** . Any other data should be **rejected**:

Test number	Data entered (test mark)	Type of test data	Expected outcome	Actual outcome
1	20	Normal	Accept	This section would be filled in by the tester and will log what actually happened as a result of each test.
2	45	Normal	Accept	
3	59	Normal	Accept	
4	17	Extreme	Accept	
5	70	Extreme	Accept	

6	Thirty	Abnormal	Reject (error message)	If any of the actual outcomes were different to the expected outcome, the module would have to be corrected and then re-tested .
7	16	Abnormal	Reject (error message)	
8	71	Abnormal	Reject (error message)	
9	-10	Abnormal	Reject (error message)	



Test	Test Data	Expected Results	Actual Results	Comment
Input hours worked	-1	Rejected	Error message	Abnormal Data
	65	Accepted	Wages Calculated	Extreme Data
	0	Accepted	Wages calculated but the result was 0	Extreme Data
	40	Accepted	Wages calculated	Normal Data
	140	Rejected	Error message	Abnormal Data
	Akhbar	Rejected	Error message	Abnormal Data

System modules should be tested using a testing plan with normal, extreme and abnormal test data.



Any module errors found during testing need to be corrected and then retested.

Testing the system as a whole

- # Once individual parts of the system have been tested and any problems solved, the system will be **tested as a whole**.

Testing the system as a whole is known as **integration testing**.

- # Testing the system as a whole makes sure that all of the individual modules **work with each other** correctly.

For example:

A data entry form might allow a user to enter their date of birth correctly but when submitted, this data might not be visible on the report.

- # Any **problems** found during whole system testing will be **corrected** and then **re-tested**.

Activity!

[The Testing Stage - Research Sheet](#)

Click the above task and answer all of the questions about the Testing Stage.

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Integrated systems testing is where the system is tested as a whole, across the entire organisation.



Fully tested systems are more likely to work properly and lead to a more productive workplace.

Links to Theory Units:

[Section 1: Types and Components of Computer Systems](#)

Section 4: Networks and the Effects of using them

[Section 7: The Systems Life Cycle](#)

Section 10: Communication

[Section 2: Input and Output Devices](#)
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